Notes: Acemoglu and Yared (AER P&P, 2010)

Political Limits to Globalization

Contents

1 Big picture

- **Question.** Is globalization mechanically driven by technology and economic efficiency, or can political forces place limits on openness to international trade?
- **Core idea.** Globalization is not irreversible. Even when information technology, transportation, and global production networks make trade easier, actual openness remains a political choice made by nation states.
- **Historical motivation.** The nineteenth-century wave of globalization ended after the rise of nationalism, militarism, international conflict, and the disruption surrounding World War I and the Great Depression. The paper asks whether similar political forces may limit the contemporary wave of globalization.
- **Main proxy.** The authors use *military spending* and *military personnel* as proxies for nationalist and militarist sentiment. The interpretation is not that every dollar of military spending is nationalism; rather, military buildup is treated as an observable signal of a more nationalist, militarized political equilibrium.
- **Main empirical claim.** Countries that become more militarized experience smaller increases in trade openness. The same is true for countries whose geographically proximate trading partners become more militarized.
- **Two layers of evidence.**
  - **Country-level evidence:** long-difference regressions relate changes in a country's trade share of GDP to changes in own or neighbor militarization.
  - **Bilateral evidence:** bilateral trade between two countries grows less when the country pair becomes jointly more militarized.
- **Main quantitative result.** In the baseline country-level regression, the coefficient on the change in log military expenditure is -17.585 with robust standard error 6.296. A 10 percent increase in military spending over the period is therefore associated with roughly a 1.8 percentage point smaller increase in trade share of GDP.
- **Robustness.** The negative relationship survives excluding Asia, excluding countries involved in civil or international war, adding an OECD differential trend, using military personnel instead of military spending, and using militarization of nearby trading partners.
- **Identification status.** The paper is deliberately cautious: the evidence is not a clean causal estimate. Trade and militarization may jointly affect each other, and both may be driven by third factors. The paper's contribution is to document a robust descriptive relationship and to highlight a political-economy constraint on globalization.
- **Economic takeaway.** Globalization depends not only on technology and markets but also on political equilibria. If nationalism and militarism rise, countries may become less open even when trade remains technologically feasible and economically attractive.

2 Data and measurement (key objects)

2.1 Main outcome: trade openness

The main country-level outcome is a country's trade share of GDP:

$$y_{i,t} = \frac{\text{Trade}_{i,t}}{\text{GDP}_{i,t}}, \tag{1}$$

where $\text{Trade}_{i,t}$ is total trade and $\text{GDP}_{i,t}$ is GDP.

Data source. Trade share of GDP, real GDP per capita, and population come from the Summers-Heston dataset, 2002.

Interpretation. A higher $y_{i,t}$ means country i is more open to international trade. The regressions focus on whether this openness rises more slowly in countries that become more militarized.

2.2 Bilateral trade object

For countries i and j in year s, let

$$X_{ij,s} = \text{exports from } i \text{ to } j + \text{exports from } j \text{ to } i. \tag{2}$$

The paper constructs $X_{ij,s}$ from IMF *Direction of Trade Statistics*. Flows are measured using either FOB exports or CIF imports. When both measures are available, the authors average them; otherwise they use the available measure.

The bilateral trade share of country i's GDP is constructed as

$$y_{ij,s} = \left(\frac{X_{ij,s}}{\sum_{k \neq i} X_{ik,s}} \right) y_{i,s}. \tag{3}$$

Interpretation.

- The first term is the share of country i's total trade accounted for by partner j.
- Multiplying by i's aggregate trade share converts this into bilateral trade with j as a fraction of country i's GDP.
- This makes bilateral trade comparable across countries with different economy sizes.

2.3 Main explanatory variables: militarization

The paper uses two measures of militarization:

$$m_{i,t}^{spend} = \log(\text{military expenditure}_{i,t}), \tag{4}$$
$$m_{i,t}^{size} = \log(\text{military personnel}_{i,t}). \tag{5}$$

Data source. Military spending as a fraction of GDP and military personnel come from the Stockholm International Peace Research Institute FIRST database. The authors convert military spending into total spending using GDP per capita and population.

Interpretation.

- **Military spending** captures the resources devoted to the military.
- **Military size** captures the number of military personnel.
- Both are imperfect but observable proxies for nationalist and militarist political sentiment.

2.4 Militarization of neighbors and trading partners

The paper also constructs a measure of militarization among country i's geographically proximate neighbors. Let d_{ij} be the distance between the capital cities of i and j. Define inverse-distance weights:

$$\omega_{ij} = \frac{d_{ij}^{-1}}{\sum_{k \neq i} d_{ik}^{-1}}, \qquad j \neq i. \tag{6}$$

Neighbor militarization is

$$m_{i,t}^{N} = \sum_{j \neq i} \omega_{ij} m_{j,t}. \tag{7}$$

Interpretation.

- Nearby countries receive larger weights.
- The measure captures militarization of countries that may pose the most direct security threat to i.
- It also roughly captures militarization of countries with which i is likely to trade heavily under standard gravity logic, since nearby countries tend to be natural trading partners.

2.5 Sample periods

- **Military spending regressions:** 1988 to 2005.
- **Military size regressions:** 1985 to 2003.
- These endpoints are chosen to maximize sample size given data availability.

2.6 War and conflict exclusion

As a robustness check, the paper excludes countries that experience civil or international war between 1985 and 2005. Conflict data come from the International Peace Research Institute and Uppsala Conflict Data Program Armed Conflict Dataset.

Interpretation. This matters because the authors do not want the results to be driven only by the mechanical disruption of trade during wars. The key question is whether militarization is associated with lower trade even outside actual war episodes.

3 Empirical specifications (all equations + interpretation)

3.1 Country-level long-difference regression

The main specification is

$$\Delta y_i = \alpha \Delta m_i + \Delta x_i' \beta + u_i, \tag{8}$$

where:

- Δy_i is the change in the trade share of GDP for country i;
- Δm_i is the change in militarization, either own militarization or neighbor militarization;
- Δx_i includes changes in log GDP per capita, log population, and in some specifications an OECD differential trend;
- u_i is an error term.

Interpretation. The coefficient α asks whether countries with larger increases in militarization also have smaller increases in trade openness, relative to other countries over the same period.

3.2 Why long differences?

The long-difference form compares each country at the beginning and end of the sample:

$$\Delta y_i = y_{i,T} - y_{i,0}, \qquad \Delta m_i = m_{i,T} - m_{i,0}. \tag{9}$$

Interpretation.

- The goal is to capture medium-run changes, not year-to-year noise.
- Country-specific time-invariant factors are differenced out.
- With only two observations per country, this is algebraically similar to a panel regression with country fixed effects and a common time effect.

3.3 Panel-equivalent intuition

A corresponding two-period fixed-effect representation is

$$y_{i,t} = \mu_i + \tau_t + \alpha m_{i,t} + x_{i,t}' \beta + \varepsilon_{i,t}. \tag{10}$$

Interpretation.

- μ_i absorbs time-invariant country characteristics, such as geography or long-run institutions.
- τ_t absorbs the global trend toward greater trade.
- Identification comes from whether a country becomes more or less open than the global trend as its militarization changes.

3.4 Bilateral trade specification

The paper also studies bilateral trade with a specification of the form

$$\Delta y_{ij} = \gamma \Delta \left(m_i m_j \right) + \lambda_i + \lambda_j + \varepsilon_{ij}, \tag{11}$$

where Δy_{ij} is the change in bilateral trade between home country i and partner j as a share of i's GDP, and $\Delta(m_i m_j)$ is the change in the interaction of log militarization in the two countries.

Interpretation.

- The bilateral regression asks whether trade between two specific countries grows less when both countries become more militarized.
- Home-country and partner-country dummies absorb country-specific trends.
- Long differencing removes time-invariant country-pair characteristics.
- A negative γ means joint militarization is associated with weaker bilateral trade growth.

3.5 Coefficient interpretation

In the baseline country-level regression, the estimated coefficient on log military spending is

$$\widehat{\alpha} = -17.585. \tag{12}$$

For a 10 percent increase in military spending, $\Delta m_i \approx 0.10$, so the implied change in trade share is

$$\Delta y_i \approx -17.585 \times 0.10 = -1.76. \tag{13}$$

Interpretation. A country whose military spending rises by about 10 percent over the period has, other things equal, a trade share of GDP about 1.8 percentage points lower than it otherwise would have been.

4 Identification and credibility (what makes the estimates informative)

4.1 What identifies the relationship?

The main variation is cross-country variation in medium-run changes:

- some countries become more militarized than others;
- some countries' neighbors become more militarized than others;
- some countries experience faster or slower growth in trade openness than the global trend.

The paper asks whether these differences line up in a systematic way.

4.2 What the long-difference design removes

- **Time-invariant geography.** Distance from markets, size of the country, and other fixed geographic facts are differenced out.
- **Persistent institutional differences.** Long-run country characteristics that do not change over the sample are differenced out.
- **Global trade trend.** Because the world became more open on average, the regression effectively asks which countries rose more or less than this common trend.

4.3 Why neighbor militarization is useful

Own military spending may be jointly determined with a country's trade policy. Neighbor militarization is not a perfect instrument, but it provides a complementary source of variation:

- if nearby countries militarize, domestic politics may become more security-oriented;
- security tensions can make trade relationships more fragile;
- because the weights are distance-based, the variable is tied to geography and gravity-style trade relationships.

4.4 Robustness checks

The paper checks whether the relationship survives several alternative samples and specifications:

- **Excluding Asia.** Some Asian countries experienced both rapid militarization and rapid trade growth for reasons such as offshoring and global production shifts. Excluding Asia makes the estimated relationship more negative.
- **Excluding countries at war.** This addresses the concern that results are driven by direct wartime trade disruption. The negative relationship remains.
- **OECD differential trend.** OECD countries may have different trade-growth paths. Allowing a separate OECD trend does not eliminate the main effect.
- **Military size instead of military spending.** The results are broadly similar using military personnel.
- **Bilateral trade.** Country-pair evidence shows that joint militarization is associated with weaker bilateral trade growth, especially when militarization is measured by spending.

4.5 Main limitations

- **No clean causal identification.** The paper explicitly states that it cannot determine causality.
- **Reverse causality.** Lower trade might increase nationalism or militarization, rather than militarization reducing trade.
- **Omitted shocks.** A third factor, such as geopolitical risk, state capacity, ideology, or regional conflict risk, may affect both trade and military spending.

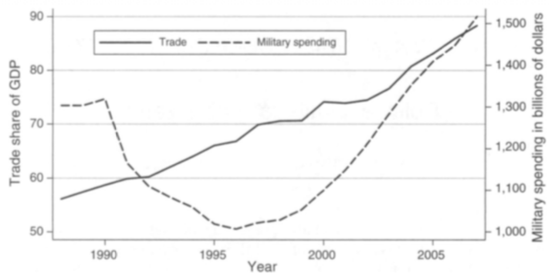

FIGURE 1. WORLD TRADE AND MILITARIZATION, 1988–2007

Notes: Sample is a balanced panel 1988–2007 for which trade share of GDP and military spending is available for all years. Trade share of GDP is the average across countries for each year. Military spending is the sum across countries and is in 1996 dollars. For reasons of data availability, military spending for Russia and China in 1989 is assigned to 1988; 1991 military spending for Russia is interpolated; and 1990 openness for Russia is assigned to 1988 and 1989. See text for data definitions and sources.

- **Proxy problem.** Military spending is an imperfect measure of nationalism. It can reflect external threats, alliance commitments, technology, income, or defense modernization.
- **Short article.** This is an AER Papers and Proceedings article, so the empirical analysis is intentionally compact. Some bilateral details are deferred to the working paper.

Bottom line. The design is best read as strong descriptive evidence, not as a definitive causal estimate. Its value is to discipline a broad political-economy claim with cross-country and bilateral facts.

5 Tables and figures

5.1 Figure 1: World Trade and Militarization, 1988–2007

Read: The figure plots average world trade share of GDP and total military spending for a balanced panel of countries from 1988 to 2007.

- Trade rises steadily over the period.
- Military spending falls through the early to mid-1990s, then begins rising again.
- The figure motivates the paper: globalization and renewed militarization can coexist at the aggregate level, so the question is whether cross-country changes reveal political limits to trade.

Interpretation. The aggregate trend does not by itself prove that militarization lowers trade. Instead, it shows that rising global trade does not eliminate nationalist or militarist forces. The regressions then ask whether countries with larger militarization increases have weaker trade growth.

5.2 Table 1, Panel A: military expenditure

Read: Panel A uses log military expenditure as the militarization measure. The dependent variable is the change in trade share of GDP.

	Base	Excl. Asia	Excl. war	OECD trend	Both variables
Own log military expenditure	−17.585 (6.296)	−22.562 (7.053)	−24.529 (9.261)	−17.144 (6.072)	−15.342 (6.555)
Neighbor log military expenditure	−33.032 (14.259)	−36.852 (15.155)	−28.506 (16.121)	−28.745 (13.157)	−21.736 (14.568)

Key numbers.

- Baseline own militarization: −17.585 with standard error 6.296.
- Excluding Asia makes the coefficient more negative: −22.562.
- Excluding countries at war also makes it more negative: −24.529.
- Neighbor militarization is also negative: baseline −33.032.
- When own and neighbor military expenditure are included together, own militarization remains negative and statistically important; neighbor militarization remains negative but is no longer precisely estimated.

Interpretation. Countries with greater increases in military expenditure experience weaker increases in trade openness. This pattern is not driven only by Asia, OECD status, or direct war disruptions.

5.3 Table 1, Panel B: military size

Read: Panel B repeats the exercise using log military personnel instead of military expenditure.

	Base	Excl. Asia	Excl. war	OECD trend	Both variables
Own log military size	−10.006 (4.356)	−11.426 (4.797)	−8.831 (6.151)	−9.939 (4.295)	−7.241 (3.887)
Neighbor log military size	−59.256 (16.028)	−48.796 (13.563)	−51.713 (14.304)	−55.671 (16.094)	−53.396 (15.644)

Key numbers.

- Own military size is negatively associated with trade openness: baseline −10.006.
- Neighbor military size is strongly negative: baseline −59.256.
- In the combined specification, neighbor military size remains large and negative: −53.396.

Interpretation. The basic pattern is not unique to spending. It also appears when militarization is measured by the number of military personnel. Neighbor militarization is especially strong in the military-size specifications.

5.4 Figure 2: Change in Trade and Militarization, 1988–2005

Read: Figure 2 is a residual plot corresponding to column 1 of Panel A in Table 1.

- The horizontal axis is the residual change in log military expenditure.
- The vertical axis is the residual change in trade share of GDP.

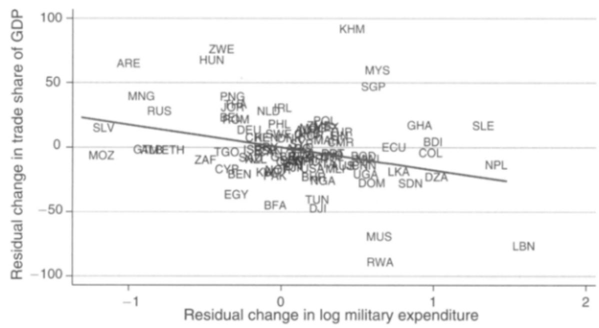

FIGURE 2. CHANGE IN TRADE AND MILITARIZATION, 1988–2005

Notes: Figure corresponds to the residual plot of regression in column 1 of panel A of Table 1. See text for data definitions and sources. coef. = −17.585122, (robust) s.e. = 6.2961775, $t = -2.79$.

- The regression line slopes downward.
- The reported coefficient is -17.585, robust standard error 6.296, and $t = -2.79$.

Interpretation. After removing the effects of changes in GDP per capita and population, countries with larger increases in military spending have smaller increases in trade openness.

5.5 Bilateral trade evidence

Read: The paper also studies whether trade between two countries changes when the country pair becomes jointly more militarized.

- The dependent variable is the change in bilateral trade between home country i and partner j as a fraction of home-country GDP.
- The key regressor is the change in the interaction of the two countries' log militarization measures.
- The specification includes home-country and partner-country dummies.
- The paper reports that the interaction effect is negative, statistically significant, and economically large when militarization is measured by military spending.
- The results are weaker when militarization is measured by military size.

Interpretation. The bilateral results support the country-level evidence. Trade links appear to weaken when both sides of the relationship become more militarized.

6 What the paper contributes to the literature

6.1 Economic history of de-globalization

The paper connects to work on why the nineteenth-century globalization wave ended. The key point is that economic integration can be reversed by political forces, including nationalism, militarism, and international conflict.

6.2 Liberal peace literature

The international-relations literature often studies whether trade reduces war. This paper looks at the reverse or complementary channel: militarization and nationalist sentiment may reduce trade.

6.3 Trade attitudes and nationalism

The paper also relates to evidence that nationalist sentiment is negatively associated with support for trade. Its contribution is to move from survey attitudes to country-level and bilateral trade outcomes.

7 Short summary

- Globalization is not automatic; it depends on political decisions by nation states.
- The paper uses military spending and military personnel as proxies for nationalism and militarism.
- The main country-level regression is a long-difference OLS regression of change in trade share of GDP on change in militarization and controls.
- The baseline coefficient on own log military expenditure is -17.585 with robust standard error 6.296.
- Excluding Asia, excluding war countries, and adding an OECD trend do not remove the negative relationship.
- Neighbor militarization is also negatively associated with trade growth.
- Bilateral trade evidence suggests that country pairs trade less when both countries become more militarized.
- The evidence is not causal, but it is consistent with the idea that nationalism and militarism impose political limits on globalization.

8 Conclusion

8.1 Core conclusion

Acemoglu and Yared argue that openness to trade is a political choice, not an inevitable consequence of technology. Their evidence shows a robust negative association between militarization and trade openness during the late twentieth and early twenty-first centuries.

8.2 Economic intuition

Militarization can shift political priorities away from gains from trade and toward security, sovereignty, and national rivalry. These forces may weaken support for open borders, reduce trust between trading partners, or make governments more willing to restrict trade.

8.3 Takeaway

The main lesson is that globalization has political limits. Even when markets and technology favor integration, rising nationalism and militarism can slow or reverse openness. The paper is therefore less about estimating a final causal parameter and more about warning that the political foundations of globalization cannot be taken for granted.